

Module -1

DC Circuit Analysis and Theorems

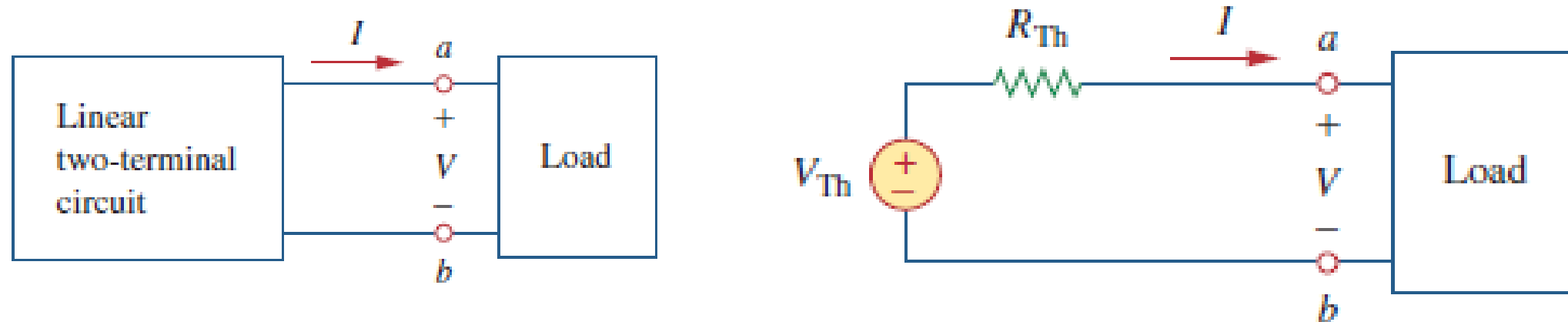
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SENSE

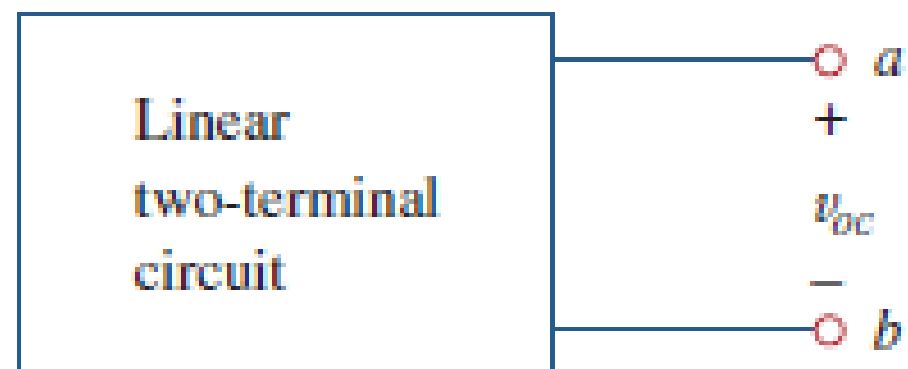
Thevenin's Theorem

- In a circuit, a particular element is variable while other elements are fixed.
- Eg. A household outlet terminal may be connected to different appliances constituting a variable load.
- Each time the variable element is changed, the entire circuit has to be analysed all over again.
- To avoid this problem, Thevenin's theorem provides a technique by which the fixed part of the circuit is replaced by an equivalent circuit.

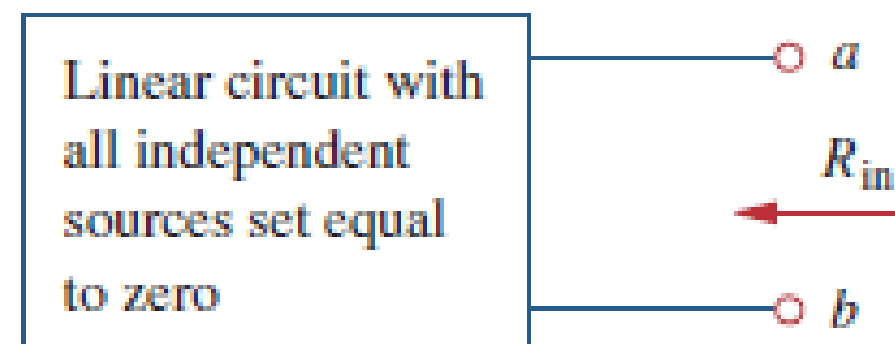
Thevenin's Theorem

Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

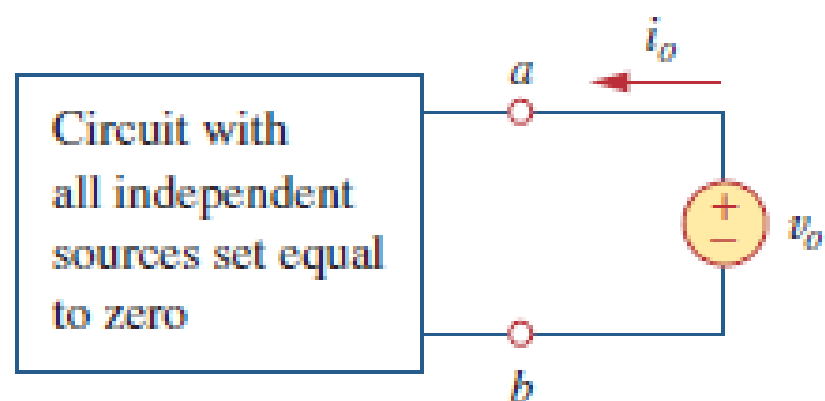




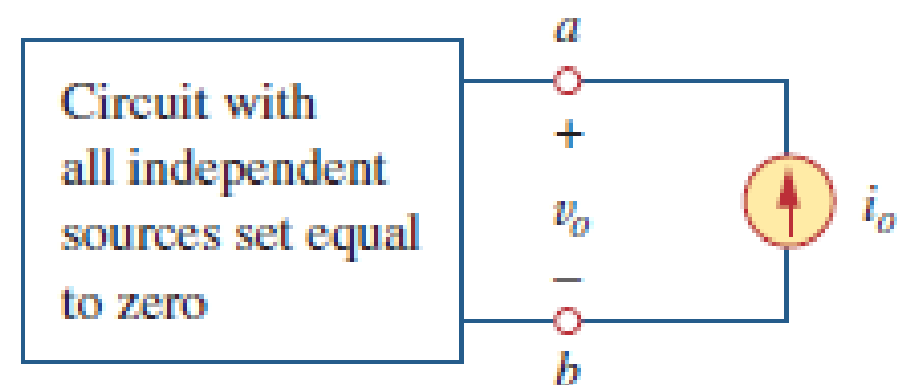
$$V_{Th} = v_{oc}$$



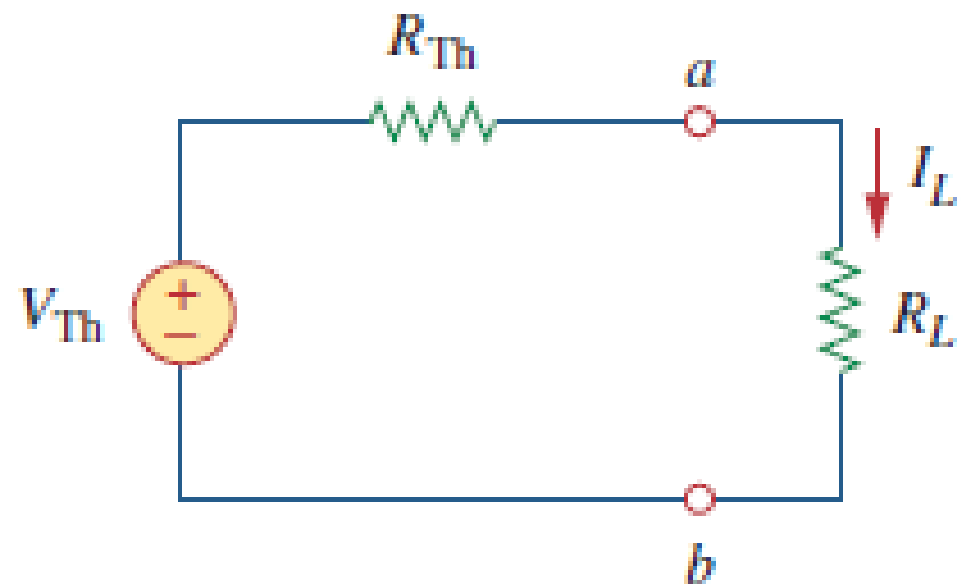
$$R_{Th} = R_{in}$$



$$R_{Th} = \frac{v_o}{i_o}$$



$$R_{Th} = \frac{v_o}{i_o}$$

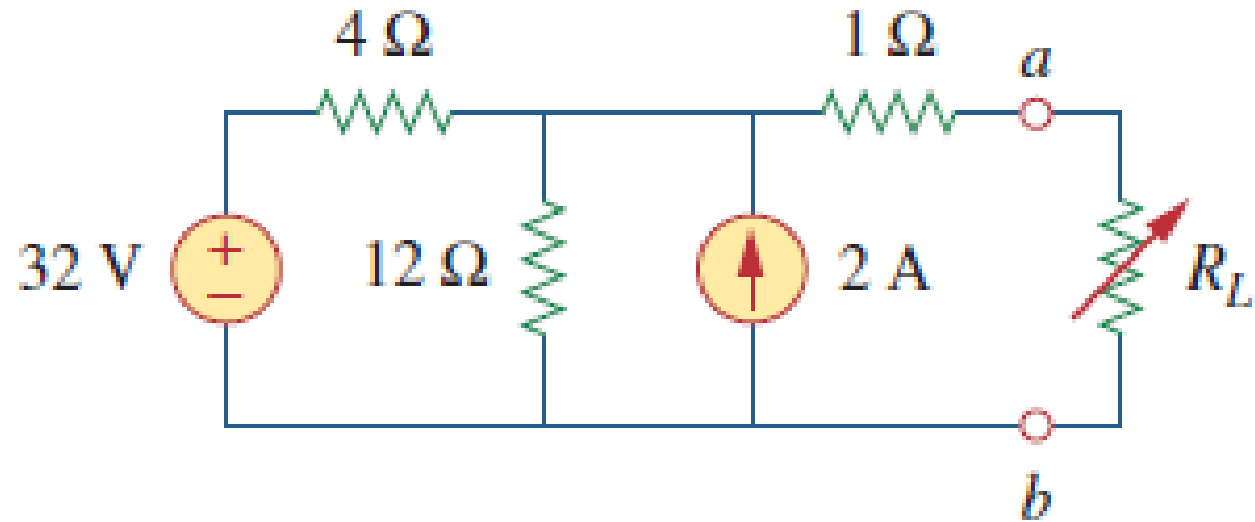


$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

Tutorial - 1

Find the Thevenin equivalent circuit of the circuit shown to the left of the terminals a-b. Find the current through $R_L = 6, 16$ and 36Ω .



Solution:

$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4 \Omega$$

$$V_{Th} = 30 \text{ V}$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

When $R_L = 6$,

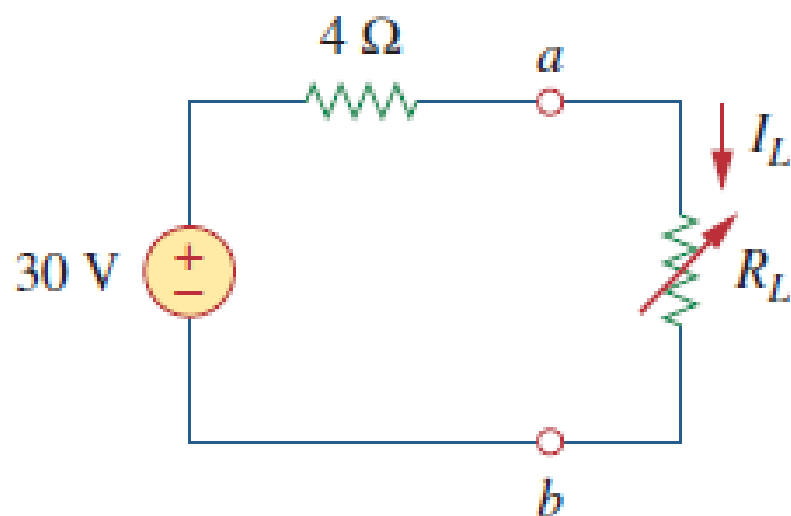
$$I_L = \frac{30}{10} = 3 \text{ A}$$

When $R_L = 16$,

$$I_L = \frac{30}{20} = 1.5 \text{ A}$$

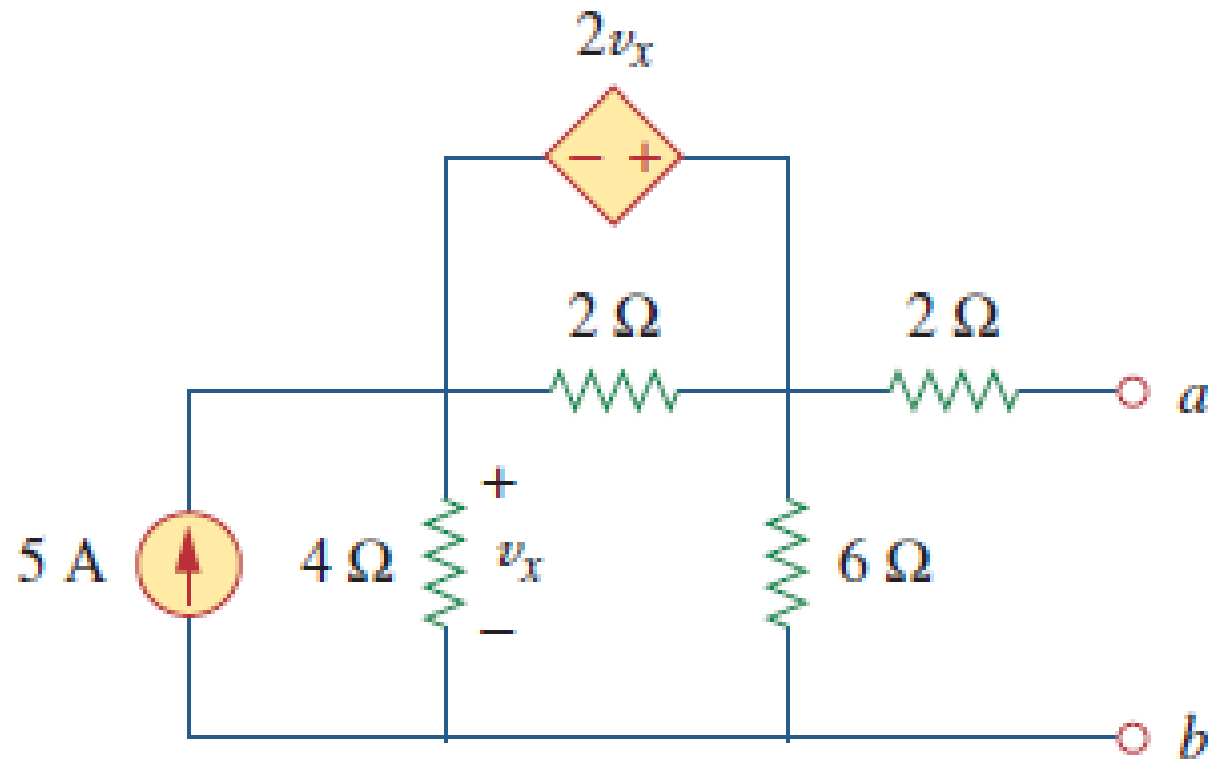
When $R_L = 36$,

$$I_L = \frac{30}{40} = 0.75 \text{ A}$$

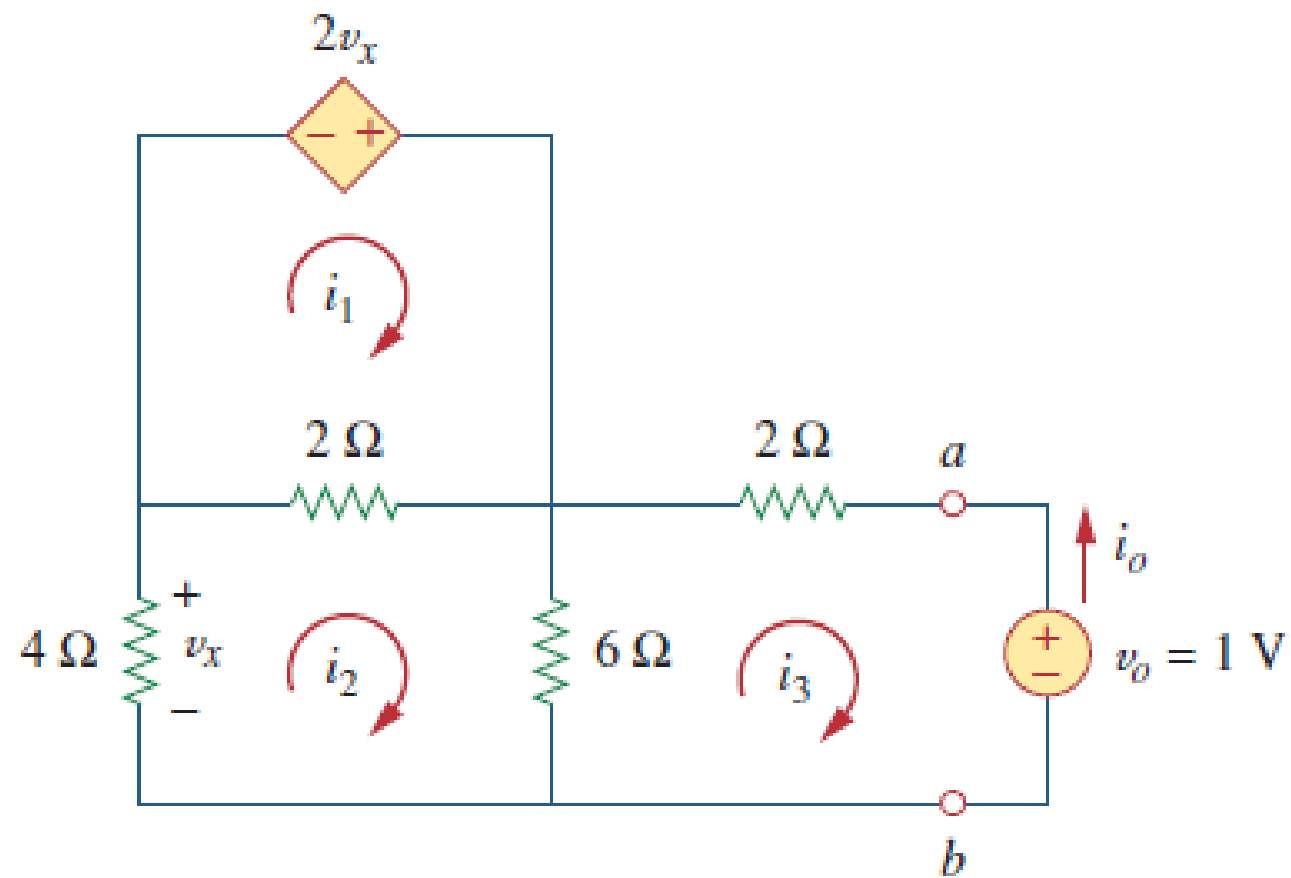


Tutorial - 2

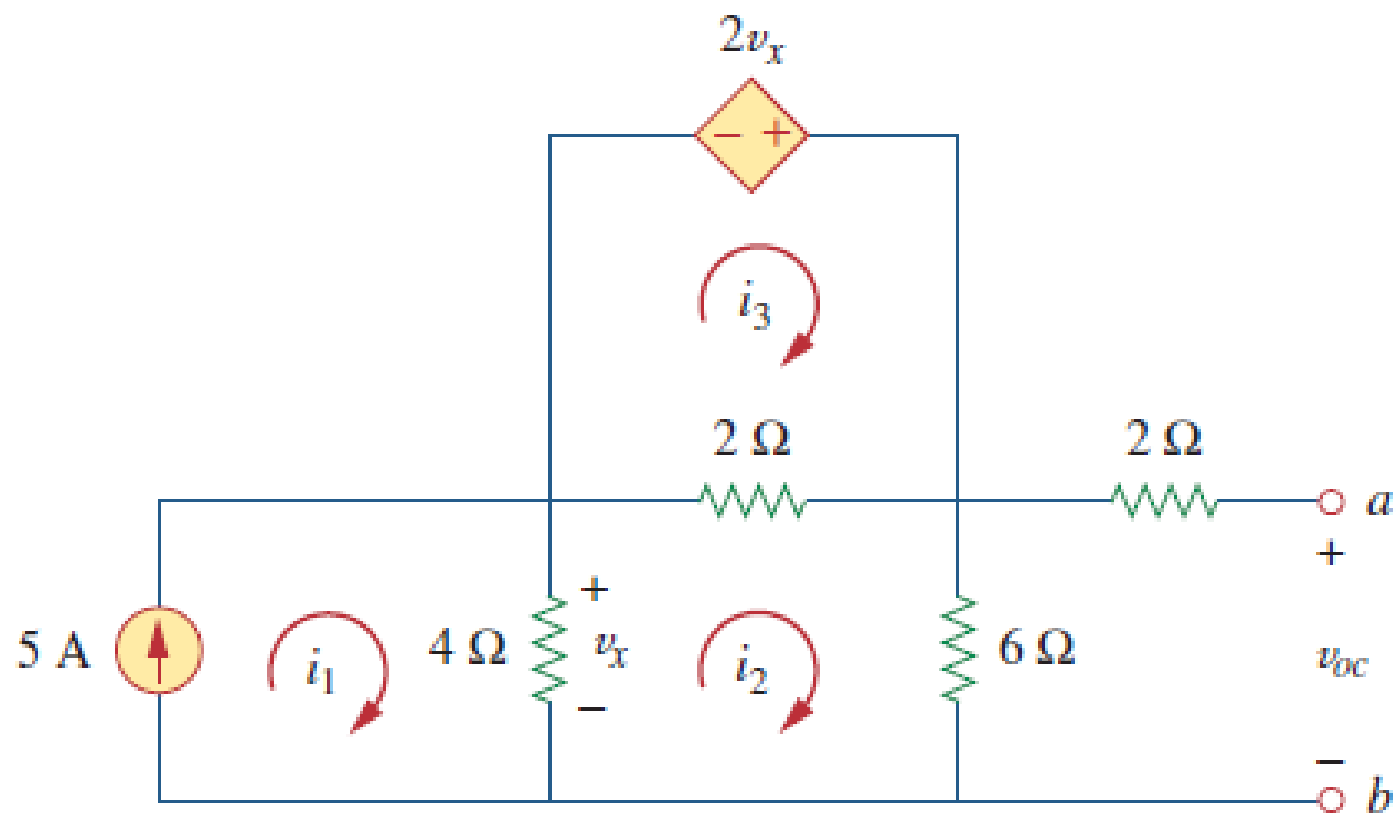
Find the Thevenin equivalent of the circuit at the terminals a-b.



Solution:



$$R_{\text{Th}} = \frac{1\ \text{V}}{i_o} = 6\ \Omega$$



$$V_{Th} = v_{oc} = 6i_2 = 20 \text{ V}$$